Middlesex University Dubai

CW1 PART 2: EPSON Robot Pick-and-Place Programming Task

PDE4431 – Robot Manipulation

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1. Coursework Objectives

1.1 Aim of the Coursework

The coursework under Robotic Manipulation (PDE4431) aims to develop and implement a robot program on the EPSON VT6/SCARA robot to perform an automated pick-and-place operation. The task involves programming the robot to pick up an object from a designated position, transport it safely, and place it accurately at multiple target locations following the required placement pattern. Through this exercise, the objective of the course, as a whole, is to understand the practical application of industrial robot programming via EPSON RC+. Learning to define robot positions, controlling gripper, selecting appropriate motion commands and implementing tool offsets are all part of understanding how automated pick-and-place sequences are structured, tested, developed and optimised in an industrial robotics environment and constitutes the aims of the coursework.

1.2 Task Requirements

The task requires the development of a functional pick-and-place operation using the EPSON RC+ version 8.0 environment. The robot must begin its operation from a defined home reference position and execute the required sequence of movements to pick up and place objects at the designated locations as follows:

- Establishing communication between the robot controller and the programming computer via the USB cable.
- Configuring and verifying the Tool Centre Point (TCP) for the gripper (end of arm tooling).
- Programming the robot using appropriate motion commands, including GO and MOVE.
- Implementing approach and retract points to ensure safe access to the pick-up and drop positions, along with implementing continuous paths where necessary.
- Controlling the gripper through both the code and the robot's digital outputs to securely pick up and release objects and ensuring that it gets enough time to open or close before moving.
- Applying tool offsets, particularly along the Z-axis, to achieve approach heights.
- Using coordinate offsets or position-shifting techniques to generate the required placement pattern.
- Testing both the individual movements and the complete pick-and-place sequence to assess movement accuracy, placement consistency, and overall operation.
- Ensuring that the robot returns to its home position after completing the sequence.

2. System Setup and Configuration

2.1 Robot, Programming Environment and Assigned Task

The pick and place operation was carried out using the EPSON VT6-A901S robot, manufactured by Seiko Epson Corporation, specifications of which can be found in Table 1, programmed via the EPSON RC+ 8 programming environment using the SPEL+ programming language. (Smart Programming and Execution Language Plus). [1]

Specification	**Details**
Manufacturer	EPSON
Robot model	VT6-A901S
Robot type	Industrial robot
Maximum reach	1000 mm
Maximum payload	6 kg
Robot weight	40 kg
Power supply	AC 100–240 V
Frequency	50/60 Hz
Rated current	12 A
Axis 1 motor power	300 W
Axis 2 motor power	300 W
Axis 3 motor power	200 W

Axis 4 motor power	100 W
Axis 5 motor power	100 W
Axis 6 motor power	100 W
Manufacturing date	27 June 2023

Table 1 – Specifications of EPSON VT6-A901S



Figure 1- Epson VT6

The EPSON RC + software gives users the ability to define robot positions, configure movement instructions and coordinate robot operations with external equipment such as grippers. The robot's six-axis configuration allows flexibility in movement across different positions and orientations, which makes it especially suitable for industrial automation applications. [2]

The task assigned using the robot was to pick ping pong balls from a stand and place them onto the assigned positions on the metal plate as shown in Figure 2.



Figure 2 - Assigned Drop Positions for Pick and Drop Operation

The program is divided into separate functions to make the entire operation easier to organise and understand. This structure allows the robot to perform the required sequence without placing all the instructions into one long program. The functions will be further discussed in Section 4.

2.2 System Start-Up and Controller-to-PC Communication

Before executing the program, the robot system and programming environment were prepared for operation with the robot controller being powered on, and the EPSON RC+ software being opened to access the project and robot program.

Communication between the robot controller and the programming laptop is required, to transfer and execute programs through EPSON RC+, which is done via the USB port with the setting changed in the task bar as shown in Figure 3. This allows the programmed movements and gripper operations to be tested using the physical robot as the software provides the interface through which the SPEL+ program can be edited, compiled and executed on the robot.

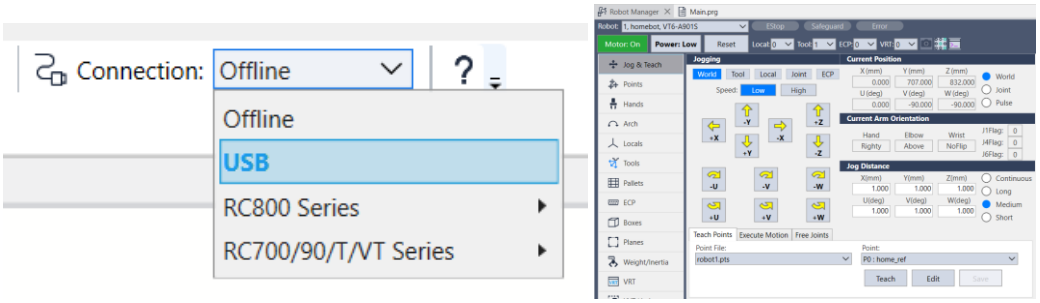


Figure 3 – USB Port Connection and Jog and Teach Options

To confirm the connection, the robot can be moved in the Robot Manager using the controls in the Jog and Teach segment under the World Frame, as demonstrated in Figure 3, once the motor is turned on. Another point to note is to avoid configurations where Joints 5 and 4 align, as this can result in a robot singularity and cause restricted robot movement.

Once the robot's connection is properly established, the tool centre point needs to be set, which will be detailed in Section 3, after which points must be taught and run.

The taught positions, including the home/reference position, pick-up position and placement positions, must be used as the movement targets in the program. The gripper operation must also be checked to ensure that it could open and close through the assigned digital output.

The program must be compiled before execution to identify any syntax or programming errors. Once the program is ready, the robot could be operated through the controller to test the pick-and-place sequence.

2.4 Preparation of a Personal Reference Sheet

As part of the preparation for the practical task, a personal reference sheet, as displayed in Figure 4, was created to assist with robot programming and the use of the EPSON RC+ environment. The sheet was handmade and organised as a quick reference guide to make important programming instructions, commands and concepts easier to access during development.

Additionally, comments for the different sections of the robot program were prepared beforehand to serve as a reference during coding. These comments, which are attached below, helped organise the program structure and provided guidance on the purpose and sequence of the instructions being implemented. Both resources were used as supporting references during the development of the final SPEL+ program.

```
Function main
  'Call REQUIRED PICK DROP FUNCTIONS
Fend

Function Grip_open
    --------
Fend
Function Grip_close
    ------
Fend
Function Pick
    -------------------------
```

```
    'Open gripper at start for ease for closed gripper: ON 11
    'Can create a function called Gripper open or gripper close to make it work
    ---------
    'Introduce a delay so that the gripper can open in time
    --------
    'Approach Point of Pick
    --------------------
    'Pickup Point
    --------------------
    'Wait for Robot to reach point then close gripper
    ------
    ------
    'Wait for gripper to actually take the object
    --------
    'Move to Retract Point
    ------------------
Fend

Function Drop
    'Approach Point of Drop
    -----------------
    'Drop Point
    --------------
    'Wait for Robot to reach point then open gripper
    ------
    ------
    'Wait for gripper to actually drop the object
    ------
    'Move to Retract Point
    --------------------
'Move to home_pt
    --------------------
Fend
```

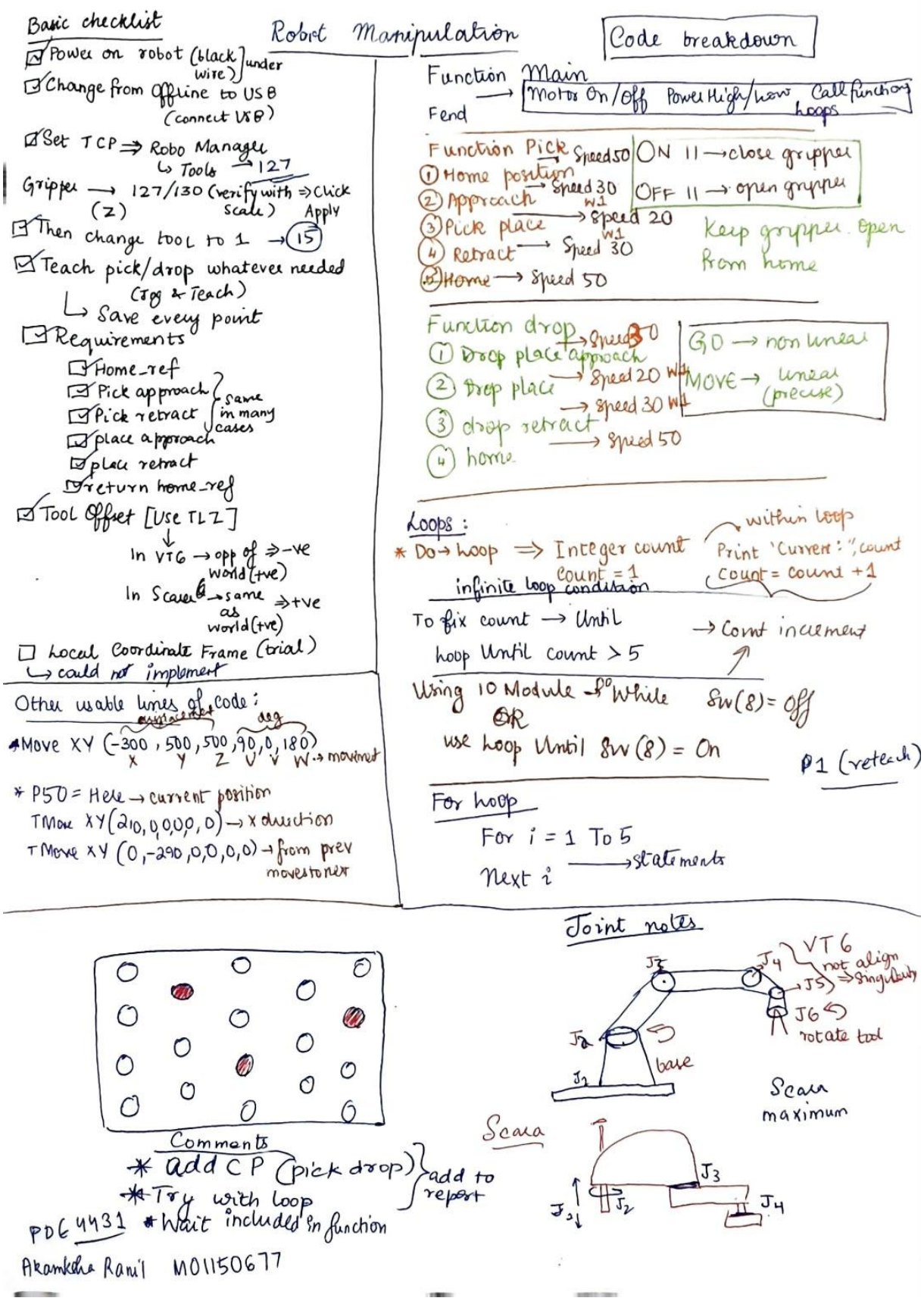


Figure 4 – Personal Reference Sheet

3. TCP, Tool Configuration and Teaching of Points

3.1 Tool Centre Point (TCP)

The Tool Centre Point (TCP) is the reference point used by the robot to determine the position of its end-effector during movement. In a pick-and-place operation, it is important that the TCP corresponds to the intended gripping point of the tool, as this affects how accurately the robot approaches, picks up and releases an object. An incorrectly defined TCP can cause the gripping point to deviate from the intended position, affecting pickup and placement accuracy. [3]

3.2 TCP Configuration and Verification

The TCP is configured in EPSON RC+ using the appropriate tool configuration procedure. For this task, the robot was equipped with an end-effector used to grip the object during the pick-and-place sequence. The gripper used here, due to its structure, has an invisible TCP, which was entered using the direct entry method as 127 mm. [3]

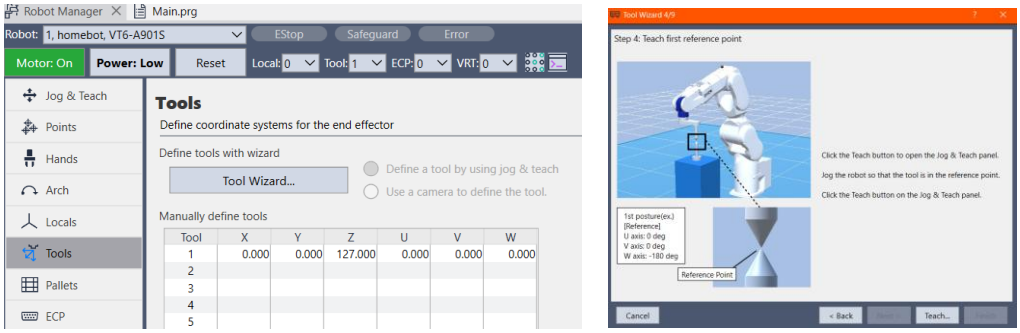


Figure 5 – Tool Centre Point

A more accurate method of finding the TCP is using the Tool Wizard, which is visible in Figure 5. This wizard guides the user through the TCP calibration process by using multiple taught positions and different robot orientations to determine the tool centre point. This helps establish the TCP more accurately than manually entering an estimated value. To perform this procedure, the Tool Wizard is opened from the tool configuration options. The robot is then positioned so that the tool tip touches a fixed reference point, and multiple positions are recorded by changing the robot's orientation while keeping the tool tip at the same point. The application uses these recorded positions to calculate the TCP coordinates, which can then be saved and assigned to the robot tool. [4]

3.3 End-Effector and Gripper Configuration

The end-effector used for the task was a gripper controlled through digital output channel 11, either directly through the I/O Monitor as seen in Figure 6, or via the code. Two separate functions were created in the program to control its operation:

- ***Grip_open*** uses Off 11 to open the gripper.
- ***Grip_close*** uses On 11 to close the gripper.

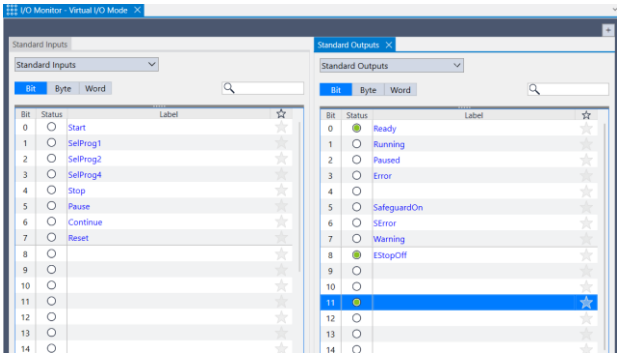
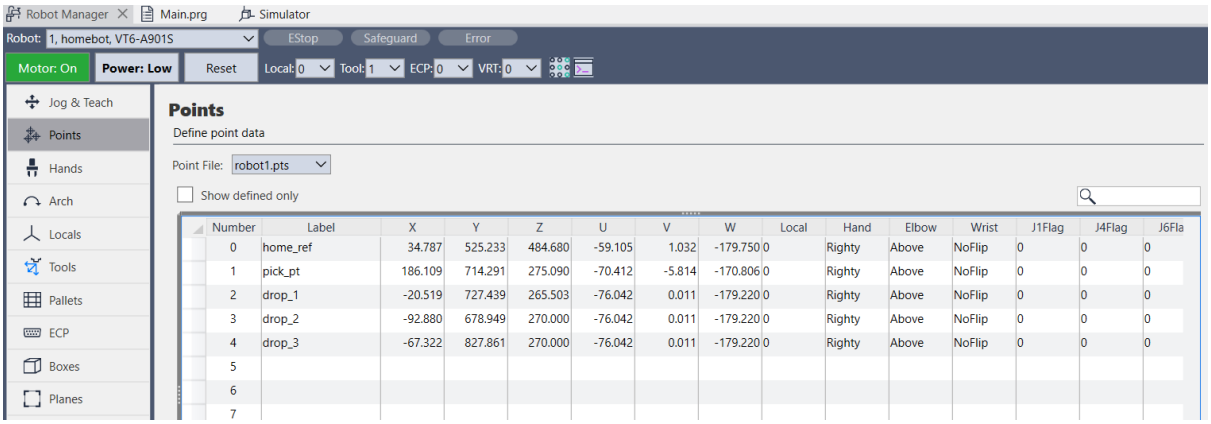


Figure 6 – I/O Monitor

The gripper is opened before the robot approaches the pick-up position and closed after the robot reaches the object. At each drop position, the gripper is opened to release the object. This allows the same gripper control functions to be reused throughout the program.

3.4 Teaching of the Points

The required points for the pick-and-place operation were taught using the EPSON RC+ environment. Each point was saved after being taught and could be viewed under the Points section in the Robot Manager. These points could also be selected and edited directly if any adjustments were needed, making it easier to fine-tune the robot's positions during programming, as shown in Figure 7.



Number	Label	X	Y	Z	U	V	W	Local	Hand	Elbow	Wrist	J1Flag	J4Flag	J6Fla
0	home_ref	34.787	525.233	484.680	-59.105	1.032	-179.750	0	Righty	Above	NoFlip	0	0	0
1	pick_pt	186.109	714.291	275.090	-70.412	-5.814	-170.806	0	Righty	Above	NoFlip	0	0	0
2	drop_1	-20.519	727.439	265.503	-76.042	0.011	-179.220	0	Righty	Above	NoFlip	0	0	0
3	drop_2	-92.880	678.949	270.000	-76.042	0.011	-179.220	0	Righty	Above	NoFlip	0	0	0
4	drop_3	-67.322	827.861	270.000	-76.042	0.011	-179.220	0	Righty	Above	NoFlip	0	0	0
5														
6														
7														

Figure 7 – Taught Points for Navigation

While teaching some of the points, adjustments to the robot's orientation were also necessary to ensure proper alignment with the objects and surfaces. For example, the pick-up position had to be slightly tilted to match the angle of the inclined stand from which the ping-pong balls were picked. This was achieved using a combination of movements in the World Frame as well the Joint Frame, especially in Joint 4 and 5, which essentially helped the gripper align properly with the balls during the pick-up process.

4. Program Structure and Logic

4.1 Overview of the Program

The robot operation program is divided into multiple functions, with each function responsible for a specific part of the pick-and-place operation. [4]

The ***main*** program controls the overall sequence, while separate functions handle gripper operation, object pickup and placement. This structure makes the program easier to modify and troubleshoot.

The functions used in the program, given below are:

- ***main*** – controls the order in which the operations are performed.
- ***Grip_open*** – opens the gripper.
- ***Grip_close*** – closes the gripper.
- ***Pick*** – moves the robot to the pick-up position and grips the object.
- ***Drop1***, ***Drop2*** and ***Drop3*** – move the object to the three designated placement positions.

The code is attached below:

```
Function main
```

```
   Motor On
   Power High
   'Calling functions
   Call Pick
   Call Drop1
   Call Pick
   Call Drop2
   Call Pick
   Call Drop3
   Power Low
   Motor Off
Fend

Function Grip_open
   Off 11
Fend
Function Grip_close
   On 11
Fend
Function Pick
   'Open gripper at start for ease for closed gripper: ON 11
   Call Grip_open
   'Introduce a delay so that the gripper can open in time
   Go home_ref
   'Approach Point of Pick
   Speed 30
   Go pick_pt -TLZ(50)
   Wait 1
   'Pickup Point
   Speed 20
   Move pick_pt
   'Wait for Robot to reach point then close gripper
   Wait 1
   Call Grip_close
   'Wait for gripper to actually take the object
   Wait 1
   'Move to Retract Point
   Speed 30
   Move pick_pt -TLZ(50)

Fend

Function Drop1
   'Approach Point of Drop
   Go drop_1 -TLZ(50)
   'Drop Point
   Speed 20
   Move drop_1
```

```
   'Wait for Robot to reach point then open gripper
   Wait 1
   Call Grip_open
   'Wait for gripper to actually drop the object
   Wait 1
   'Move to Retract Point
   Move drop_1 -TLZ(50)

Fend
Function Drop2
   'Approach Point of Drop
   Go drop_2 -TLZ(50)
   'Drop Point
   Speed 20
   Move drop_2
   'Wait for Robot to reach point then open gripper
   Wait 1
   Call Grip_open
   'Wait for gripper to actually drop the object
   Wait 1
   'Move to Retract Point
   Move drop_2 -TLZ(50)
Fend
Function Drop3
      'Approach Point of Drop
   Go drop_3 -TLZ(50)
   'Drop Point
   Speed 20
   Move drop_3
   'Wait for Robot to reach point then open gripper
   Wait 1
   Call Grip_open
   'Wait for gripper to actually drop the object
   Wait 1
   'Move to Retract Point
   Move drop_3 -TLZ(50)
'Go to home_pt
   Go home_ref
Fend
```

4.2 Main Function and Execution Sequence

The main function acts as the starting point of the program and determines the order of execution. It first enables the robot motor and sets the power level to high. It then calls the pick and drop functions in the required order.

The execution sequence is:

1. From home position, go to pick the object from the designated pick-up position.
2. Place the object at the first drop position using Drop1.

3. Return to the pick function to collect the next object.
4. Place the second object using Drop2.
5. Repeat the pickup operation.
6. Place the third object using Drop3 and return to home position.
7. Reduce the power setting and switch off the robot motor.



Figure 8 – Sequence of actions in Main Function

The robot performs three consecutive pickup and placement operations.

4.3 Gripper Open and Close Functions

Two functions are created to control the gripper through digital output 11. The ***Grip_open*** function switches the output off, while ***Grip_close*** switches it on.

```
Function Grip_open
   Off 11
Fend
Function Grip_close
   On 11
Fend
```

The gripper is opened before pickup, closed once the robot reaches the pick-up point, and opened again when the object reaches its designated placement position.

4.4 Pick Function

The ***Pick*** function controls the movement of the robot towards the object and the gripping operation. It begins by opening the gripper and moving the robot to the home reference position.

The robot then moves to an approach position above the pick-up point using the tool Z-axis offset - TLZ(50). This creates a 50 mm offset along the negative tool Z direction relative to the taught pick-up position.

After reaching the approach position, the robot uses a linear MOVE command to reach the actual pick-up point. A delay is introduced before the gripper closes, allowing the robot to reach the target position. Another delay is included to allow the gripper to secure the object.

Finally, the robot moves back to the approach position, creating clearance between the object and the work surface before travelling to the placement location.

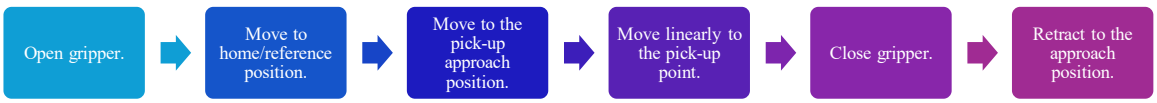


Figure 9 – Sequence of Actions in Pick Function

4.5 Drop Functions

The program contains three separate placement functions: ***Drop1***, ***Drop2*** and ***Drop3***. Each function is responsible for moving the object to a specific taught placement position.

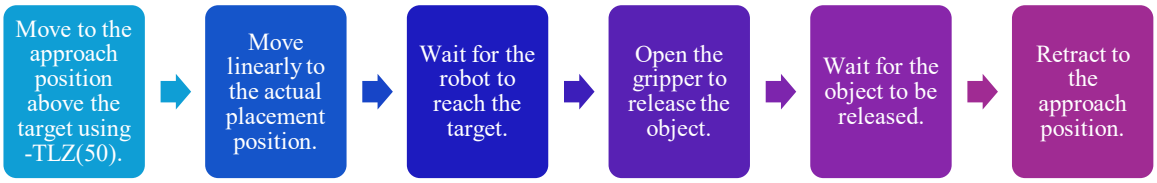


Figure 10 – Sequence of Actions in Drop Functions

The three functions use the same basic movement sequence but have different target positions: *drop_1*, *drop_2* and *drop_3*.

An additional instruction is included at the end of ***Drop3*** to move the robot back to *home_ref*. This ensures that the robot returns to its reference position after completing the final placement.

4.6 Code Modifications

The original program was structured to perform the required three pick-and-place operations. Further modifications were considered to improve the organisation and optimise the code.

4.6.1 Use of Increment Operation

Instead of repeated calling of the Pick and Drop functions in the main function, a much better and optimised way to call the functions is using a For Loop, where the Pick loop is called once every time the loop is iterated as shown below:

```
Function main
   Integer i
   Motor On
   Power High
   'Calling functions using increment operation
   For i = 1 To 3
      Call Pick
      If i = 1 Then
         Call Drop1
      Else
         If i = 2 Then
            Call Drop2
         Else
            Call Drop3
         EndIf
      EndIf
   Next i

   Power Low
   Motor Off
Fend
```

Another solution to obtain the same result is by implementing an increment operation to select the taught drop positions. Instead of creating separate functions for each drop location, a single Drop function is used along with a counter variable within a For loop.

The counter starts at 1 and increments by 1 after each iteration. The expression P(I + 1) is used to access the corresponding taught point, allowing the robot to move sequentially between P2, P3 and P4. The same approach, placement and retract movements are performed for each position, while the destination changes automatically based on the counter value. Given below is the code for the same:

```
Function main
   Integer i

   Motor On
   Power High

   For i = 1 To 3
      Call Pick
      Call Drop
   Next i

   Go home_ref
   Power Low
   Motor Off
Fend


Function Drop
   'Approach the matching drop point
   Go P(i + 1) -TLZ(50)

   'Move to the drop point
   Speed 20
   Move P(i + 1)

   'Open gripper to release object
   Wait 1
   Call Grip_open
   Wait 1

   'Retract from the drop point
   Move P(i + 1) -TLZ(50)
Fend
```

4.6.2 Gripper Timing and Wait Instructions

Wait instructions are used to introduce delays during robot operation. In the original program, Wait 1 is placed after the robot reaches the pick-up or placement position. These delays allow time for the gripper to complete its operation and for the object to be securely picked up or released.

An improvement is to include the wait instruction directly inside the gripper functions. This ensures that a delay is automatically applied every time the gripper is opened or closed as shown below:

```
Function Grip_open
   Off 11
   Wait 1
Fend
Function Grip_close
   On 11
   Wait 1
Fend
```

The Wait statements in the original code defined under the Pick and Drop operations can be removed in this case. This reduces repeated wait instructions throughout the program and makes the gripper functions more self-contained.

If the delays are moved into these functions, the corresponding duplicate waits after the function calls should be removed unless an additional delay is specifically required.

4.6.3 Continuous Path

Continuous Path (CP) can be considered to improve the smoothness of the robot's movement during the pick-and-place operation. It is particularly useful during the transfer from the pickup position towards the drop location, where unnecessary stops between movements can increase the overall cycle time. CP allows consecutive robot movements to be executed with smoother transitions, reducing unnecessary pauses while maintaining the required placement accuracy. The following statements are the first of each of the Drop functions, each being called once the gripper has reached the Pick Retract Point, ensuring a smooth movement to the Drop Approach Points:

```
Go drop_1 -TLZ(50) CP
Go drop_2 -TLZ(50) CP
Go drop_3 -TLZ(50) CP
```

5. Robot Motion Commands

5.1 GO Command

The GO command is used to move the robot to a specified taught position using point-to-point movement. [5] In the program, it is used to move the robot to the home position and to the approach positions before pickup and placement.

For example:

```
Go home_ref
Go pick_pt -TLZ(50)
```

The first instruction moves the robot to the home position. The second moves it to an approach position offset from the taught pick-up point.

The GO command is useful for travelling between positions in a non-linear fashion where a direct point-to-point movement is appropriate. It is particularly useful for moving to the approach positions from the home position before performing the more controlled linear movement towards the object.

5.2 MOVE Command

The MOVE command is used to move the robot along a linear path towards a specified position. Unlike point-to-point movement, it provides a straight-line path for the robot's tool between the current position and the target. [5]

In the program, MOVE is used when approaching the actual pick-up and placement points.

For example:

```
Move pick_pt
Move drop_1
```

The robot first reaches the approach position using GO and then uses MOVE to travel linearly to the target. This helps provide a more controlled approach when gripping or releasing the object.

The MOVE command is also used to retract the robot from the pick-up and placement positions back to their respective approach positions.

5.3 Additional Commands

In addition to GO and MOVE commands, there are the CIRCLE and JUMP commands, which are not used in the current pick-and-place program. [5]

The CIRCLE command is generally used when a robot needs to follow a circular or curved path through specified points. It may be useful in applications requiring curved movement, but it is not necessary for the straightforward pickup and placement movements used in this task.

The JUMP command provides a specialised movement instruction that can be used for movements involving an approach and departure path.

5.4 Movement Speed and Selection

Movement speed determines how quickly the robot travels between programmed positions. Selecting suitable speeds is important for balancing cycle time, movement smoothness and placement accuracy.

The program uses two speed settings:

```
Speed 30
Speed 20
```

A speed of 30 is used for the approach and retract movements in the Pick and Drop functions, while a speed of 20 is selected for the linear movement towards the pick-up and drop points.

The lower speed is used when the robot is close to the object, allowing a more controlled movement during pickup and drop. The higher speed is used for movements where greater clearance is available.

The drop functions also use a speed setting of 20 for the linear movement towards their placement points. However, the initial GO instruction in each drop function appears before this speed setting. Therefore, it inherits the active speed setting rather than explicitly selecting a new speed at the beginning of the function.

For consistency, the speed can be set at the beginning of each drop function so that the approach movement also uses an intentionally selected speed.

6. Offset and Palletizing Techniques

6.1 Tool Z-Axis Offset (TLZ)

The TLZ function is used in the program to apply an offset along the tool's Z-axis. This allows the robot to move to a position above or below a taught target without requiring a separate position to be defined.

The program uses:

```
-TLZ(50)
```

This applies a 50 mm offset in the negative direction of the tool Z-axis relative to the taught position which works exactly as

```
+Z(50)
```

which is the upward offset in the World Frame.

In the program,

```
Go pick_pt -TLZ(50)
```

Is the statement which moves the robot to an approach position offset from *pick_pt.* The robot can then move to the actual pick-up point using:

```
Move pick_pt
```

The same technique is applied to the three placement positions. Using a tool-axis offset helps establish clearance during approach and retract movements and reduces the need to teach separate approach positions manually.

The actual physical direction of the offset depends on the tool's orientation as in the case of the SCARA, the tool frame and the world frame are in the same direction as opposed to the VT6. Therefore, the direction should be verified during robot testing before executing the movement at full operating speed.

6.2 Approach and Retract Positions

Approach and retract movements are important for ensuring that the robot can reach the object without making unnecessary contact with the surrounding work surface.

In this program, approach positions are generated using the TLZ(50) offset. The robot first moves to a position 50 mm away from the target along the negative tool Z direction and then moves linearly towards the actual target.

After picking up or releasing the object, the robot moves back to the corresponding offset position. This creates clearance before travelling to another location.

The same approach and retract method is used for the pick-up position and all three placement positions.

This movement sequence can be represented as:



Figure 11 – Sequence of Actions for Approach and Retract

6.3 Palletizing Sequence and Placement Pattern

The program performs three consecutive pick-and-place operations. Each operation begins with the robot collecting an object from the same taught pick-up position and transporting it to a different placement position.

The placement sequence is:

Operation	Pickup position	Placement position
First	pick_pt	drop_1
Second	pick_pt	drop_2
Third	pick_pt	drop_3

Table 2 - Placement Sequence

The three placement targets are individually defined in the program. This allows the robot to place objects at specific taught locations.

Although the task involves multiple placement positions, the current program does not automatically generate a complete pallet grid. The placement pattern is achieved through the three separately taught drop positions. An increment-based position-shifting method could be used to extend this approach to a larger arrangement, which has been discussed in Section 4.6.1.

6.4 Local Coordinate Frames

A local coordinate frame can be used to define robot positions relative to a selected reference point rather than relying entirely on the robot's global coordinate system.

This can be useful in palletizing applications because the placement positions can be defined relative to the corner or centre of a pallet. If the pallet is moved, the reference frame can be adjusted instead of individually modifying every taught position.

A local coordinate frame was not used in the supplied program. However, it could be considered as a future improvement. The process of setting a local frame is highlighted in Figure 12.

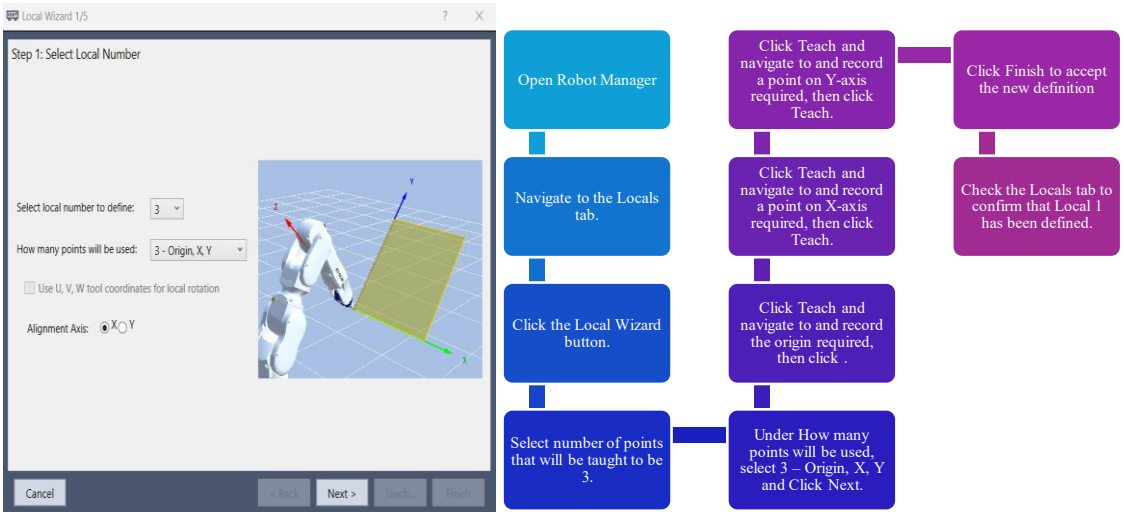


Figure 12 – Local Frame Setup

Once the local frame is configured, the required placement positions can be taught relative to the plate rather than the robot's default base coordinate system. [4]

7. Testing and Verification

Before executing the complete pick-and-place sequence, the program was compiled in EPSON RC+ to check for syntax errors and confirm that the program could be processed by the programming environment. The main function and individual subfunctions were reviewed to ensure that the required operations were called in the correct order. The taught positions and gripper output instructions were also checked as part of the initial preparation. To doublecheck the movements correctly, each point that was taught was individually executed and altered, wherever necessary. Similarly, when each pick or drop position was being taught the gripper was tested to see if it could capture and travel with the ball in the required positions.

The pick and drop operations were tested to check whether the robot could reach the designated pick-up position and operate the gripper correctly. Testing the placement positions individually helped identify issues such as incorrect taught positions, insufficient clearance or inaccurate object release.

After checking the individual operations, the main function was used to execute the complete sequence. The robot performed three pickup operations, each followed by a placement at a different drop position. The sequence was controlled through the function calls in the main program. The complete sequence was checked

to ensure that the functions executed in the intended order and that the robot could transition between pickup and placement locations.

Safety and accuracy were considered during the testing of the robot movements. The approach and retract positions were used to provide clearance around the pickup and placement targets. The 'Emergency Stop' button was kept at close proximity, in case of necessity, during the program's execution.

The use of a reduced speed near the pickup and placement points helped provide more controlled movement. Wait instructions were also included to allow time for the gripper to complete its opening and closing operations. [2]

The robot's movement and object placement were observed during testing to identify any collisions, unexpected movements or positioning errors and any necessary adjustments should be made to the taught positions, offsets or speed settings before the final demonstration.

8. Results and Observations

The developed SPEL+ program was used to execute the automated pick-and-place operation using the EPSON VT6-A901S robot. The robot successfully followed the programmed sequence, starting from the home position, moving towards the pickup location, gripping the ping-pong ball and transferring it to the assigned drop position. The sequence was repeated for all three placement locations on the steel plate. The gripper was controlled through the robot's digital output, with suitable waiting periods included to allow the opening and closing operations to complete. The robot also returned to its home position after completing the sequence.

The complete SPEL+ source code and project files can be accessed through the EPSON VT6-A901S Pick-and-Place GitHub Repository. To view the practical implementation, click on the following link to watch the Pick-and-Place Demonstration Video.

The placement accuracy depended on the taught positions, tool configuration and approach and retract heights used in the program. The use of separate taught points allowed the robot to reach the three assigned placement locations on the steel plate.

The approach and retract movements helped maintain clearance from the plate during movement to and from the placement positions. The programmed sequence also allowed the same pickup and placement operations to be repeated consistently. Any minor variations in placement could be associated with the accuracy of the taught points, TCP configuration and gripper positioning.

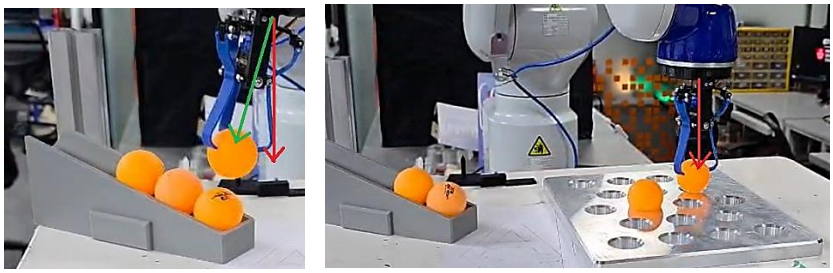


Figure 13 – Gripper Adjustments

During the development and testing of the program, adjustments were required to ensure the correct execution of the pick-and-place sequence. The robot positions and approach heights had to be checked to ensure that the gripper could reach the objects without interfering with the stand or steel plate. An initial issue encountered was that of the angle of the pick. Capturing only about 30% of the ball on the first try, resulted in it falling during travel or falling into the wrong drop location after being released. Hence why the robot was moved along the Joint Frame as well to make sure that the ball is captured as precisely as possible as demonstrated in Figure 13.

9. Conclusions and Lessons Learned

The coursework provided practical experience in developing and implementing an automated pick-and-place operation using the EPSON VT6-A901S industrial robot and EPSON RC+ programming environment. The task involved configuring the robot, defining taught positions, controlling the gripper through digital outputs and implementing appropriate motion commands to perform the required sequence. The coursework helped bridge the gap between theoretical concepts in industrial robotics and their practical application through robot programming and testing. Additionally, the SPEL+ programming threw light on several concepts that are small but very significant in automating the robot arm movements like the defined taught positions, gripper controls or motion commands, which become crucial steps in controlling the arm movement.

9.1 Limitations and Possible Improvements

During the implementation, some difficulties were encountered while teaching and adjusting the robot positions, particularly in ensuring that the gripper could reach the ping-pong balls and place them accurately without interfering with the stand or steel plate. Adjustments to the robot's joint positions and angles were also necessary to achieve the required orientation during the pickup and placement operations. As shown in Figure 13, shifting the robot through the Joint Frame helped achieve the required positioning and orientation for the pick-and-place operation.

Although the pick-and-place sequence was implemented successfully, there are several areas where the system could be improved further. The current program relies on predefined taught positions, meaning that any changes in the location or orientation of the steel plate would require the positions to be checked or retaught.

The best methods of improvement include incorporating the Local Coordinate system into the program to define placement positions relative to the steel plate, as well as simplifying it using the For loop, changes which have been highlighted in Section 4.6.

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